

## Update Hollandse Kust (west) Seabed Morphodynamics



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# Summary

Deltares updated the morphodynamic assessment (Deltares 2023) for the HKW VII Wind Farm Site at the request of JBO GmbH, incorporating new 2024 bathymetric survey data. The study aimed to (1) validate previous predictions and (2) refine the methodology and results using updated data.

The validation showed good agreement between predicted and measured seabed levels, with a Root Mean Square Error (RMSE) of 0.19 m, confirming the reliability of the previous methodology.

The seabed in and around HKW VII is dominated by sand banks and sand waves with variable migration rates and directions. Using combined large-scale bathymetry, sand wave predictions and including uncertainties, future seabed changes were projected from 2024 to 2064. The study provides a Best-Estimate Bathymetry (BEB) and Lowest and Highest SeaBed Levels (LSBL/HSBL) annually for 2024 to 2027, and at five-year intervals for 2029 to 2064. These projections offer improved insight into expected seabed conditions at anticipated project milestone dates (installation and decommissioning) and throughout the full operational lifespan of the wind farm. The influence of human interventions is not considered in this analysis. Moreover, no additional safety margins for design purposes have been applied to these bed levels.

Data and figures of the seabed level projections for the abovementioned years are available in the accompanying data package. An overview of the files in the data package is provided in Appendix A to C.

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# 1 Introduction

## 1.1 Project background

The Dutch government has designated several areas in the North Sea for the development of offshore wind farms. The Hollandse Kust (west) Wind Farm Zone, located off the Dutch coast, is one of these areas. The Hollandse Kust (west) Wind Farm Zone (hereafter HKW), is located approximately 53 kms off the west coast of the Netherlands and is divided into three Wind Farm Sites (WFS) of which two (VI and VII) are under development (Figure 1.1). HKW VII is being developed by RWE Offshore Wind and includes the development, construction and commissioning of a 760 MW Offshore Wind Farm in the Dutch Sector of the North Sea. The water depths in and around HKW VII range from 20 to 34 m MSL. The WFS will be connected to the shore by the Hollandse Kust (west Beta) offshore substation.

An initial quick scan of HKW was performed by Deltares (2017), focusing on a general overview of the area, including geology, morphology and hydrodynamics. This study was further extended by Arcadis Germany GmbH and Geo-Engineering.org GmbH (2018), where geology and morphology were investigated using data sources that were available at that time and by DHI (2019), investigating metocean conditions for HKW. In 2020 a detailed investigation of the morphodynamics for HKW was performed. That analysis consisted of a scour and scour mitigation study and a morphodynamic study by Deltares (2020). The publicly available report and data for the study by Deltares (2020) covers the period 2019 to 2059.

In 2023, JBO (Jörss-Blunck-Ordemann GmbH) requested Deltares to update the morphodynamic study for the HKW VII Wind Farm Site. The results of the 2023 study (Deltares, 2023) cover the period 2025 to 2063. More recently, JBO obtained an additional bathymetric survey for the HKW VII Wind Farm Site. This new survey forms the basis for the update presented in this report.

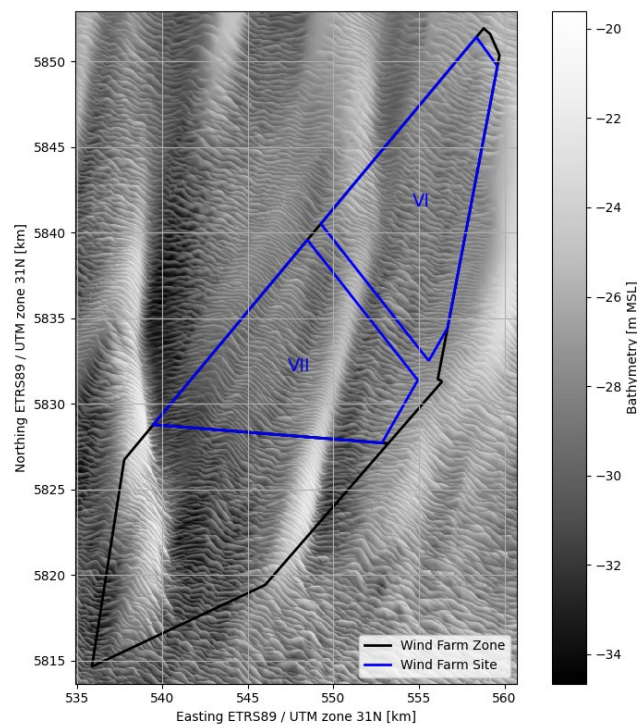


Figure 1.1: Polygons for HKW (black), HKW VI and HKW VII (blue).

## 1.2 Objectives

JBO (Jörss - Blunck - Ordemann) GmbH has requested Deltares to update the morphodynamic assessment for the HKW VII Wind Farm Site, following the acquisition of an additional bathymetric survey for the area.

The objective of this update is twofold: (1) validate the results of previous study by Deltares (2023) with the newly acquired bathymetry survey; and (2) to update the methodology, originally detailed in Deltares (2020), and results by incorporating this new survey. More specifically, the update will include:

- Best-Estimate Bathymetry (BEB) predictions annually for 2024 to 2027 and at five-year intervals for 2029 to 2064, covering to the anticipated installation and decommissioning dates of the HKW VII Wind Farm;
- Lowest SeaBed Levels (LSBL) and Highest SeaBed Levels (HLSBL) expected between 2024 and 2064, at five-year intervals, representing the full operational lifespan of the wind farm.

Note that the hydrodynamic and sediment transport modelling scope that were part of the original study by Deltares (2020) are not updated.

## 1.3 Reader

Chapter 2 provides an overview of the available data. Chapter 3 presents the validation of the new bathymetric data against the results from the previous Deltares study. Chapter 4 assesses the morphodynamics in the area, while Chapter 5 focuses on the future seabed level analysis. Finally, the conclusion is presented in Chapter 6.

The estimated future seabed levels are presented as maps within this report; the corresponding digital files (data and figures) are available in the accompanying data package. An overview of these figures is provided in Appendix A to C.

## 2 Data availability

In this chapter, the available data is presented. The scope of this study is limited to updating the methodology, originally presented in Deltares (2020), and results, from Deltares (2023), by incorporating the new 2024 bathymetric survey. Therefore, only the bathymetric data (Section 2.1) is described in this chapter. A comprehensive overview of the other data (i.e. geotechnical and hydrodynamic) can be found in Deltares (2020).

### 2.1 Bathymetric data

The two main sources of bathymetric data for this study comprise of (i) historical bathymetry data available from the Netherlands Hydrographic Office (NLHO) and (ii) data provided by JBO. An overview of the number of surveys available for the HKW is presented in Figure 2.1. The figure includes both historic seabed surveys provided by NHLO and the Client. The majority of the area is covered by at least 4 surveys that span a period of approximately 28 years. The availability of the great number of surveys improves both the temporal and spatial coverage of the analysis of seabed dynamics. The datasets are summarized in Table 2.1.

Other data sources, such as GEBCO and data from admiralty charts are not used in the analysis because of limited spatial coverage and large uncertainties in bed levels.

All available datasets are interpolated to a resolution of 5x5 m for the analysis presented in the remainder of this report. All surveys were checked for errors in the vertical reference levels, e.g. in case a survey is on average significantly deeper than all other surveys, and if required this was corrected.

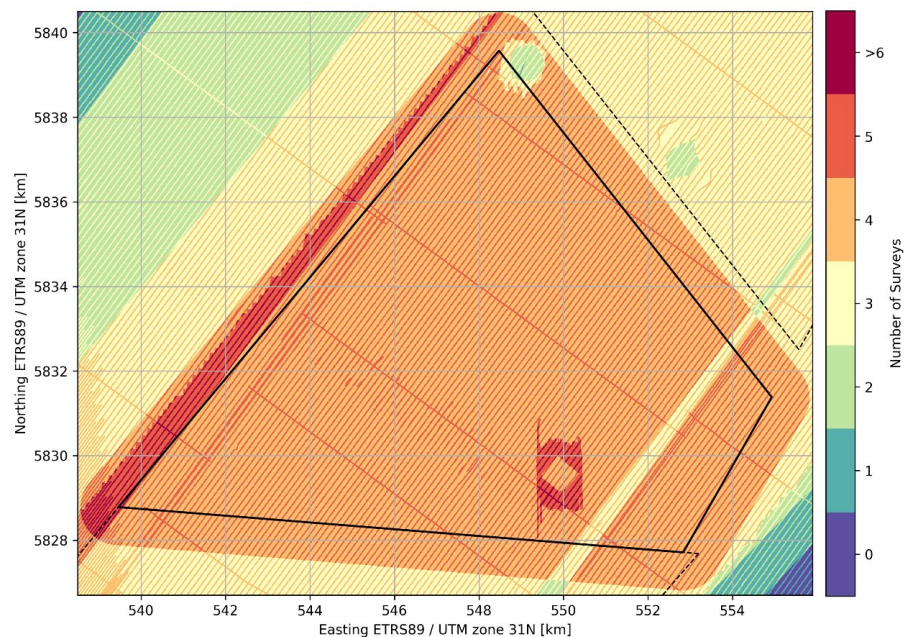


Figure 2.1: Overview of the number of surveys available for HKW VII. Note that this figure contains all available surveys from the NLHO and the Client.

Table 2.1: Available bathymetry data in and around HKW VII.

| Year      | Reference                                                     | Coverage                                                                                                                                             | Survey resolution                                           |
|-----------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1996-1997 | NHLO (Royal Netherlands Navy - Hydrographic Office 1980-2021) | Full coverage of HKW VII and the surrounding area.                                                                                                   | Single Beam Echo Sounder (SBES), lines spaced ~100 m apart. |
| 2006-2008 | NHLO (Royal Netherlands Navy - Hydrographic Office 1980-2021) | Cover the majority of HKW VII and the surrounding area.                                                                                              | MBES 5x5 m                                                  |
| 2012      | NHLO (Royal Netherlands Navy - Hydrographic Office 1980-2021) | Cover the area located directly north-east of HKW VII                                                                                                | MBES 5x5 m                                                  |
| 2014      | NHLO (Royal Netherlands Navy - Hydrographic Office 1980-2021) | Full coverage of HKW VII and partial coverage of the surrounding area.                                                                               | MBES 5x5 m                                                  |
| 2018-2019 | Fugro (Fugro 2019)                                            | Full coverage of HKW VII and partial coverage of the surrounding area.                                                                               | MBES 5x5 m                                                  |
| 2022      | TenneT (no reference)                                         | Coverage of the area surrounding the TenneT platform located within the area.                                                                        | MBES 0.5x0.5 m                                              |
| 2024      | Fugro (Fugro 2025)                                            | Full coverage of HKW VII, except for the area surrounding the TenneT platform located within the area, and partial coverage of the surrounding area. | MBES 1x1 m                                                  |

The surveys, mainly the 2024 bathymetry, contain multiple areas that are influenced by human interventions (Figure 2.2). These areas were first removed from the surveys in which these human interventions are observed. Interpolation was then applied to reduce the influence of human interventions on the results of the analysis.

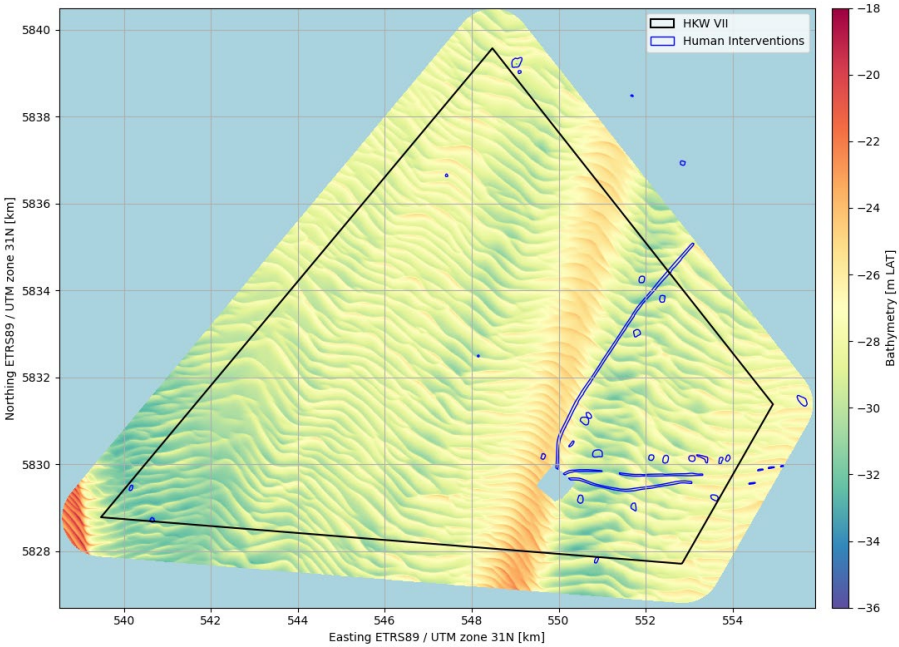


Figure 2.2: Overview of the areas influenced by human interventions (blue polygons).

## 3 Validation of previous study

In this chapter, the results of the morphodynamic study for the HKW VII Wind Farm Site by Deltares (2023) are validated by comparing them to the newly obtained bathymetric survey. This is done to validate the methodology applied in the previous study. The study presented in this report follows the same methodology as the one applied in Deltares (2020) and Deltares (2023). study. Therefore, this validation provides valuable insights that improve the understanding of the mobility in HKW VII and ensures the quality of the current study. This chapter is split into two brief sections describing the methodology (Section 3.1) and results (Section 3.2).

### 3.1 Methodology

In this validation the Best-Estimate Bathymetry (BEB), Lowest SeaBed Levels (LSBL) and Highest SeaBed Levels (HSBL) predictions for 2024 (presented in Deltares (2023) are compared to the measured 2024 bathymetry. These comparisons are made by subtracting the predicted bathymetries from the measured bathymetry. The differences are quantified using the Root Mean Square Error (RMSE) and a range of percentiles.

### 3.2 Results

Figure 3.1 presents the difference between the predicted best-estimate bathymetry for 2024 and the measured bathymetry for 2024. Red and blue patches and lines are clearly visible in this figure. These patches and lines occur due to local offsets or corrections applied during the processing of individual surveys. Such artifacts typically arise from differences in vessel positioning, tidal corrections, sensor calibration, or georeferencing between survey campaigns. As these inconsistencies are introduced during data acquisition or processing, they do not reflect actual morphological changes on the seafloor. Larger differences are observed along the North-Western border of the area of interest. This is because the best-estimate bathymetry along this stretch is determined based on an older survey (i.e. the 2014 bathymetry) compared to the rest of the area of interest (i.e. the 2018-2019 bathymetry). The largest morphodynamic differences (i.e. locally up to 2.0 m) are observed around human interventions (e.g. cable corridors and anchor pullouts). These are mainly located in the eastern part of the area (Figure 2.2). Finally, the sand wave crests can be observed in this figure. This means that the predicted and measured bathymetry do not match perfectly. Small differences between the actual sand wave migration rates and best-estimate migration rates can lead to significant differences (i.e. up to 0.5 meters) at the steep slope of the sand wave crests.

The Root Mean Square Error (RMSE) and a range of percentiles for the comparison of the predicted 2024 bathymetries and measured 2024 bathymetry are listed in Table 4.2. The RMSE between the best-estimate bathymetry and measured bathymetry is 0.19 m. This value is within the 0.20 m survey inaccuracy generally accepted for MBES surveys (International Hydrographic Organization 2020). Furthermore, the LSBL and HSBL including uncertainties are mostly respectively below or above the measured bathymetry (see 99.5% percentile in Table 3.1).

Based on the limited differences between the best-estimate bathymetry and measured bathymetry, and the measured bathymetry being mostly enclosed between the lower and upper bounds of the predicted seabed levels, it is concluded that the methodology applied in the previous study, and originally detailed in Deltares (2020), is capable of predicting the seabed levels in the area of interest.

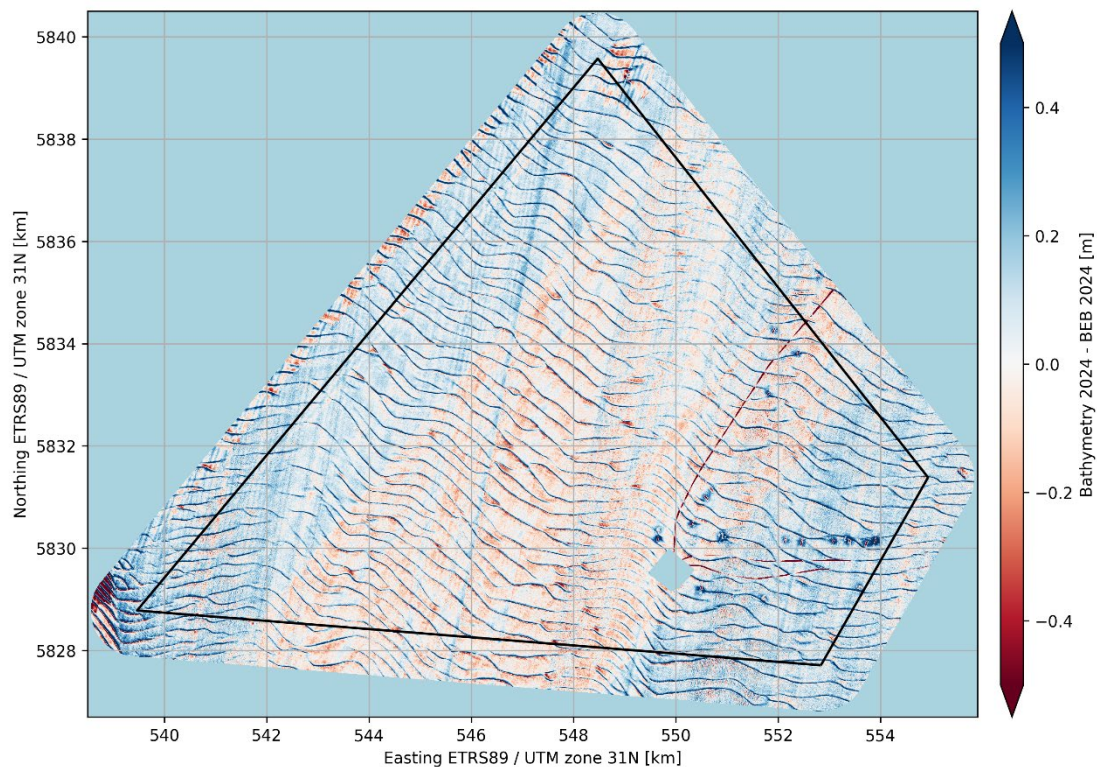


Figure 3.1: Difference between the predicted Best-Estimate Bathymetry for 2024 and the measured 2024 bathymetry.

Table 3.1: Overview of the RMSE and a range of percentiles for the comparison of the predicted 2024 bathymetries (BEB, LSBL and HSBL) and measured 2024 bathymetry.

| Predicted bathymetry               | RMSE [m] | 0.5% [m] | 1% [m] | 5% [m] | 50% [m] | 95% [m] | 99% [m] | 99.5% [m] |
|------------------------------------|----------|----------|--------|--------|---------|---------|---------|-----------|
| <b>BEB 2024</b>                    | 0.19     | -0.33    | -0.26  | -0.15  | 0.04    | 0.35    | 0.73    | 0.88      |
| <b>HSBL 2024</b>                   | 0.51     | -1.12    | -1.01  | -0.83  | -0.44   | -0.1    | 0.14    | 0.25      |
| <b>LSBL 2024</b>                   | 0.59     | -0.01    | 0.03   | 0.11   | 0.31    | 1.38    | 2.12    | 2.3       |
| <b>LSBL 2024 incl. uncertainty</b> | 0.93     | 0.4      | 0.43   | 0.52   | 0.73    | 1.78    | 2.52    | 2.7       |
| <b>HSBL 2024 incl. uncertainty</b> | 0.99     | -1.68    | -1.57  | -1.37  | -0.95   | -0.57   | -0.35   | -0.24     |

## 4 Morphodynamic analysis

In this chapter, the methodology (Section 4.1) and results (Section 4.2) of the morphodynamic analysis are presented. This analysis comprises of two steps: splitting of bedforms and analysis of sand wave dynamics. These steps follow the methodology presented in Deltares (2020).

### 4.1 Methodology

#### 4.1.1 Morphodynamic overview and splitting of bedforms

Three main bedform types are present in the area. From large to small scale, these are: sand banks, sand waves and megaripples. In general, sand banks are large ridges with wavelengths on the order of 1–10 km. They are typically relatively stable bedforms with low migration rates. Literature indicates that sand banks are often covered by sand waves. Sand waves are dynamic bedforms that generally migrate at rates of 0–20 m/year, with wavelengths ranging from 100–1000 m and heights of a few metres (Van Dijk 2011). The smallest scale bedforms considered in this study, the megaripples, are commonly superimposed on sand waves and sand banks. Typically, megaripples have wavelengths on the order of metres and heights up to approximately 1 m.

The different types of bedforms were separated to analyse them individually. Separate layers are created for the sand waves, the large-scale underlying seabed and megaripples (Table 4.1). First, the migrating part of the bathymetry has been separated from the underlying, large-scale bathymetry (the sand banks) and overall trends in the seabed. For this purpose, a 2D ellipsoidal filter was applied to the available surveys. The filter size and orientation was chosen such that the mobile bedforms (i.e. sand waves and megaripples) could be separated, while the underlying bathymetry remains unaltered in shape and is not noticeably smoothed by the filtering process. Next, a circular filter was used to split the mobile bathymetry into two components: a sand wave field and a megaripple field. Table 4.2 contains the specific filter settings used to split the bathymetry into separate layers.

The results are analysed by examining the presence of short-wavelength gradients in the underlying large-scale bathymetry, which indicate the existence of bedforms. Additionally, the mobile bathymetries are assessed to determine how much of the large-scale seabed structure remains present. An ideal mobile bathymetry should reflect only the bedforms, fluctuating around zero.

The separated layers (Table 4.1) form the basis for the analysis of sand wave dynamics and the future seabed levels.

Table 4.1: Definitions of various bathymetrical data fields used in this study.

| Short name             | Layer description                                        | Large-scale seabed | Sand waves | Megaripples |
|------------------------|----------------------------------------------------------|--------------------|------------|-------------|
| Bathymetry             | Full measured bathymetry including all bedforms          | ✓                  | ✓          | ✓           |
| Large-scale bathymetry | Filtered bathymetry with the large-scale seabed only     | ✓                  | ✗          | ✗           |
| Mobile bathymetry      | Filtered bathymetry with sand waves and megaripples only | ✗                  | ✓          | ✓           |
| Sand wave field        | Filtered bathymetry with sand waves only                 | ✗                  | ✓          | ✗           |
| Megaripple Field       | Filtered bathymetry with megaripples only                | ✗                  | ✗          | ✓           |

Table 4.2: Filters used to split the bathymetry into separate layers.

| Filter description                                           | Shape       | Orientation [°N] | Length [m] | Width [m] |
|--------------------------------------------------------------|-------------|------------------|------------|-----------|
| Split bathymetry into large-scale and mobile bathymetry      | Ellipsoidal | 10               | 1400       | 140       |
| Split mobile bathymetry into sand wave and megaripple fields | Circular    | -                | 23         | 23        |

#### 4.1.2 Analysis of sand wave dynamics

Sand waves are generally characterised by a mild sloping stoss side and a steeper lee side oriented in the direction of propagation. This distinction may be less pronounced in areas where sand waves can migrate in opposing directions. Furthermore, sand waves tend to migrate in a direction roughly perpendicular to the crest with a constant form in the direction of the steepest gradient of the sand wave stoss side or in the direction of the residual current. This may be less pronounced in areas where the underlying large-scale seabed shows a highly non-uniform morphology. This knowledge is used to analyse the sand wave dynamics and provides a good first indication of the dynamics of the seabed as a result of sand wave migration.

To determine the sand wave dynamics first the crests of the bedforms are detected. This is done using a peak detection algorithm similar to the one applied in Damen, van Dijk et al. (2018). From this, analysis locations are defined at a regular interval of 50 m along the sand wave crest. The mobile bedforms of all available surveys are analysed by comparing them with each other. Comparisons are made at all analysis locations for which more than two surveys are available. This ensures that both spatial and temporal variability are considered when evaluating the sand wave dynamics.

The sand wave dynamics are determined using a 2D cross-correlation. This technique computes local morphodynamics using the horizontal translations of bedforms from bathymetrical surveys. An example of this technique is presented in Figure 4.1, for an area around a single analysis location. The size of the considered area is chosen such that at least a significant part of one sand wave is covered in both surveys that are compared, considering the possible migration distance of the sand wave in the period between the surveys. It is also essential to limit the area to assess sand wave dynamics locally. The red arrow in the figure indicates the shift between the two surveys (black and blue polygons).

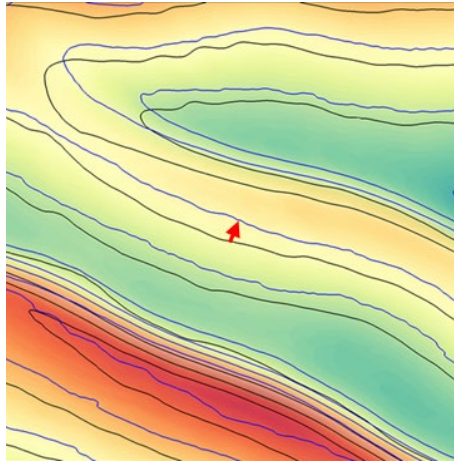


Figure 4.1: Example of the 2D cross-correlation. The red arrow indicates the highest correlation found between an older (black contour lines) and a more recent bathymetry (blue contour lines and presented as the figure background), with the arrow length indicating migration distance and the orientation migration direction.

Based on the above-presented method, the sand wave migration direction (direction of the arrow) and the sand wave migration distance (length of the arrow) are estimated. The sand wave migration rate is derived by dividing the migrated distance, as derived from the 2D cross-correlation of two bathymetries, by the difference in measurement dates of the two bathymetries.

This analysis yielded a single value for the direction and rate per analysis location per survey comparison. It is noted that no values are found if one or both surveys contain insufficient data for the specific analysis location. Local variations over time are quantified to capture the full temporal variability of sand wave dynamics at each analysis location. More specifically, the distribution of the migration directions and rates of all the neighbouring locations within a selected distance is calculated for each analysis location. The distance is selected such that the enclosed area is small enough to maintain local information, but at the same time, it should contain enough points to determine probability distributions.

Sand waves migrating in opposite directions might cause issues when determining the spread in migration directions. Averaging the migration direction for sand waves migrating in opposite directions can result in a resulting average migration direction perpendicular to the actual direction of migration. This error can occur when determining the spread in migration directions for a single analysis location or when interpolation is performed between several analysis locations.

To cover for sand waves migrating in opposing directions, first a prevailing direction of migration is determined. Since sand waves are characterised by a steeper lee slope oriented in the direction of migration, the most prevailing direction of this lee slope is considered. All sand waves migrating roughly opposite of this direction are considered to have a negative migration rate, hence the direction of migration is flipped 180 degrees. For example, when the prevailing direction of migration is 10°N, and a sand wave is migrating with 2 m/year towards 190°N, the sand wave is considered to migrate with a rate of -2 m/year towards 10°N.

Finally, minimum and maximum rates are determined at every location by taking the average rates plus or minus two times the standard deviation. Including the minimum and maximum rates reduces uncertainties in the temporal evolution of sand wave migration directions and rates. The resulting bandwidths including mean values, are used to assess future seabed levels.

## 4.2 Results

### 4.2.1 Separated bedforms

The results of the bedform separation for the most recent 2024 bathymetry are presented in Figure 4.2. The top left panel depicts the original unfiltered bathymetry, whereas the filtered large-scale bathymetry is shown in the top right panel. The large sand bank located within the area is clearly visible in the large-scale bathymetry, whilst the smaller bedforms have been removed.

The bedforms, i.e. sand waves and megaripples, obtained by subtracting the filtered large-scale bathymetry from the original unfiltered bathymetry are shown in the bottom panel. These bedforms are expressed around a zero mean. The sand waves in the area have heights between 1 and 5 meters and wavelengths between 100 and 500 m respectively. The highest sand waves within HKW VI are found to the west of the sand bank. The sand waves on top of the bank are rather small, and shorter compared to the surrounding area. Moreover, it is observed that sand waves on top of the sand bank located west of HKW VI are higher than those observed within the area. Since the megaripples are much smaller than the sand waves, they are not clearly visible in Figure 4.2. This mobile bathymetry is used for the extrapolation described in Chapter 5.

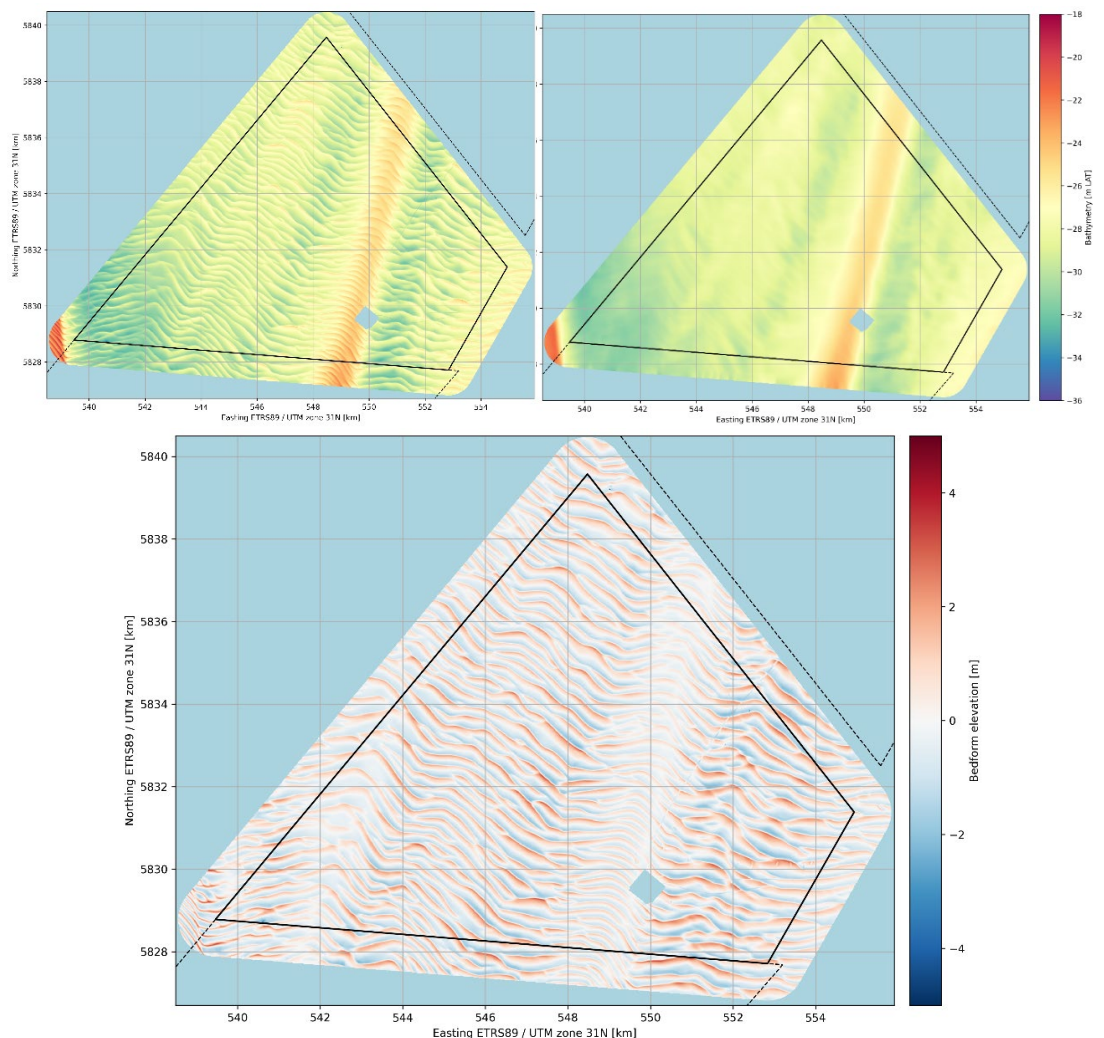


Figure 4.2: The original unfiltered bathymetry for the year 2024 (top left); the filtered large-scale seabed, where both the sand waves and megaripples are excluded (top right); the sand waves and megaripples (not clearly visible) (bottom).

#### 4.2.2 Migration rates and direction

Figure 4.3 presents the analysis locations along the sand wave crestlines for all surveys. The migration rates and directions are determined at these locations. The small offset between consecutive surveys is caused by the sand wave migration.

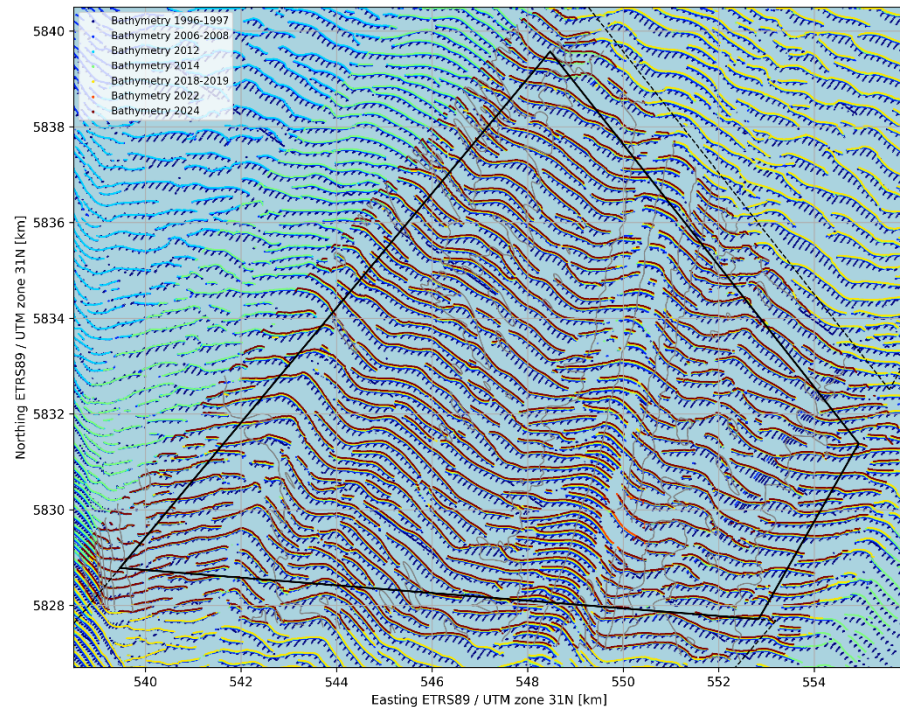


Figure 4.3 Overview of the analysis locations, located along the sand wave crestlines, for all available bathymetry surveys.

The results for the sand wave migration patterns determined using the 2D cross correlation are presented in Figure 4.4. The migration rates and directions are depicted in the top and bottom panel, respectively. The migration rates within HKW VI typically vary between 1 and 4 m/year. The influence of the large-scale bathymetry is visible as sand waves migrate faster on top of the sand banks. The directions are shown in degrees relative to the north and defined as “going towards”. A migration direction of 45°N would thus mean that the sand waves migrate towards the north-east. As seen in the figure, the sand waves are generally propagating towards the north-northeast, with a mean direction of approximately 10°N. However, locally migration directions can differ. These migration rates and directions are used for the extrapolation described in Chapter 5.

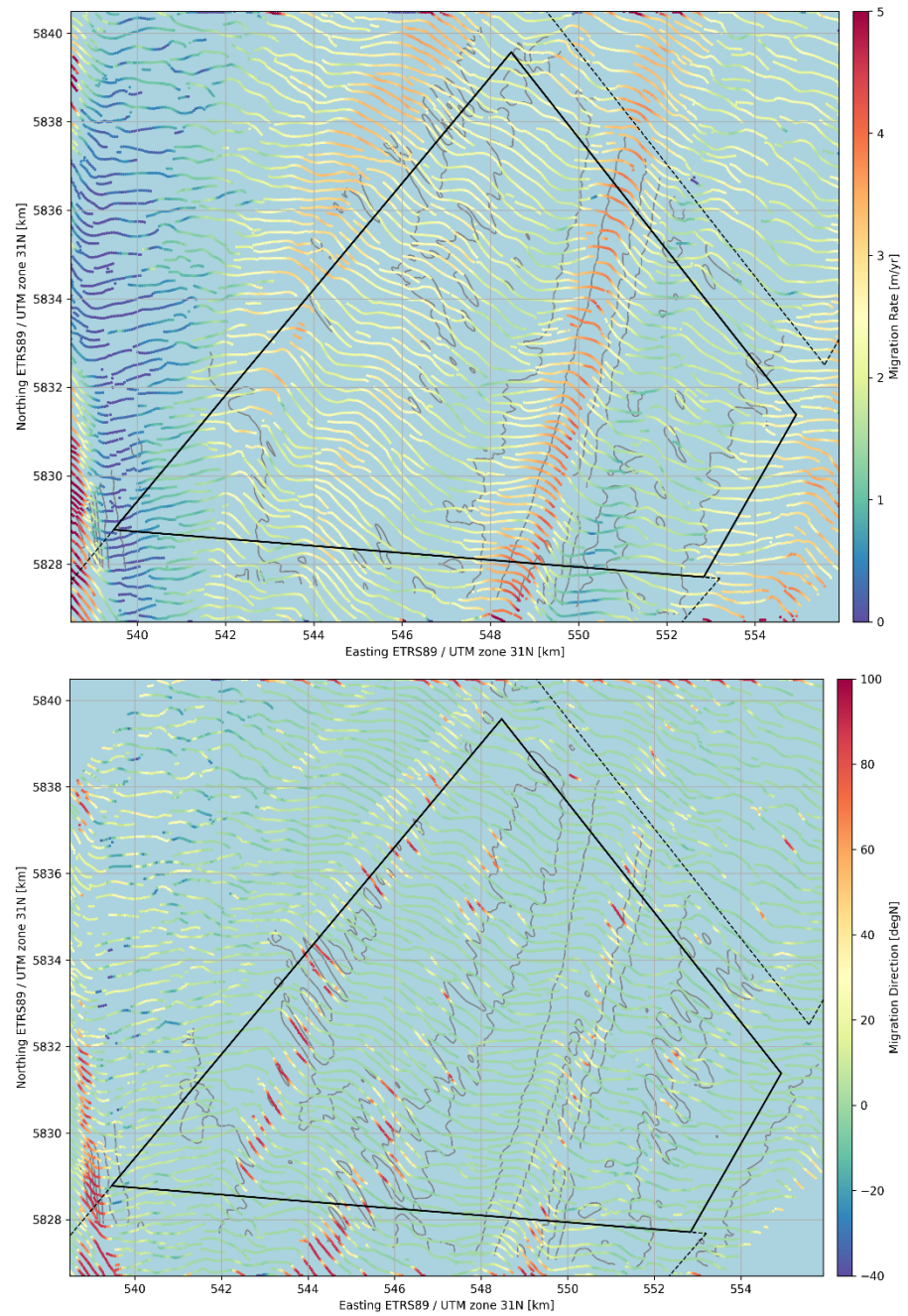


Figure 4.4 Overview of the sand wave migration rates (top panel) and migration direction (bottom panel) at all analysis locations based on comparisons between all bathymetry surveys.

## 5 Future seabed level analysis

In this chapter, the methodology (Section 5.1) and results (Section 5.2) of the future seabed levels analysis are presented. The analysis consists of three steps: extrapolating historical seabed level trends, taking into account sources of uncertainty, and assessment of future seabed levels. These steps follow the methodology presented in Deltares (2020).

### 5.1 Methodology

#### 5.1.1 Extrapolation of historic trends

The first step of the analysis consists of extrapolating historical seabed level trends. The extrapolation is performed only on the separated sand wave field. Horizontal migration of the bedforms is used for the extrapolation. Bedforms for which no historical trends can be determined, such as the highly transient megaripples, are considered as uncertainties. The extrapolation is discussed in more detail below.

##### 5.1.1.1 Extrapolation of horizontal migration of bedforms

Future bathymetries and associated bed level changes in areas with dynamic bedforms are estimated by artificially shifting the existing bedforms. The starting point for this extrapolation is a composite bathymetry representing the situation in 2024. Since the available bathymetric surveys were conducted in different years, a composite is required to ensure consistent spatial coverage. Most of the HKW VII area is covered by the 2024 survey. However, the area surrounding the TenneT platform is not included in this dataset and is instead represented by the 2022 survey. To account for the migration that has occurred within the 2 year period between these surveys, the 2022 is shifted using the mean migration rate and direction. A detailed description of this extrapolation is described below.

The artificial shifting of the sand waves is done with the aid of the determined migration directions and rates bandwidths. These bandwidths contain a lower bound, mean and upper bound for both the migration direction and rate, resulting in 9 possible displacements fully describing the temporal variation in bedform dynamics.

First, the bandwidths for all analysis locations are interpolated, resulting in bandwidths for the entire grid used in the analysis. When combining the migration directions and rates, for every grid point, the yearly shift in x and y directions is calculated. Based on the bandwidths, two approaches for the extrapolation were used, which are further explained below.

##### 5.1.1.2 Extrapolation of an individual scenario

The first approach for extrapolation consists of shifting the composite bathymetry using a single scenario, i.e. a combination of a single migration direction and rate per grid point. This approach is used to determine best-estimate scenarios consisting of the mean migration direction and rate extrapolation. An example of this approach is shown in Figure 5.1, where per grid point (black dots) the displacement in x and y directions is visualised using an arrow. This arrow indicates the direction of migration and the total displacement in that direction. This total displacement is calculated by multiplying the shift in x and y directions and the number of years for which the extrapolation is performed.

This approach results in grid points containing a certain bed level being extrapolated to a new x and y location. The resulting x, y and z locations are ultimately interpolated back to the original

grid resulting in an estimated seabed level for a specific scenario for a specific year which can be both below and above the most recent measured seabed level.

Using this method, an estimation of the bathymetry is made annually for 2024 to 2027, and at 5-year intervals for 2029 to 2064.

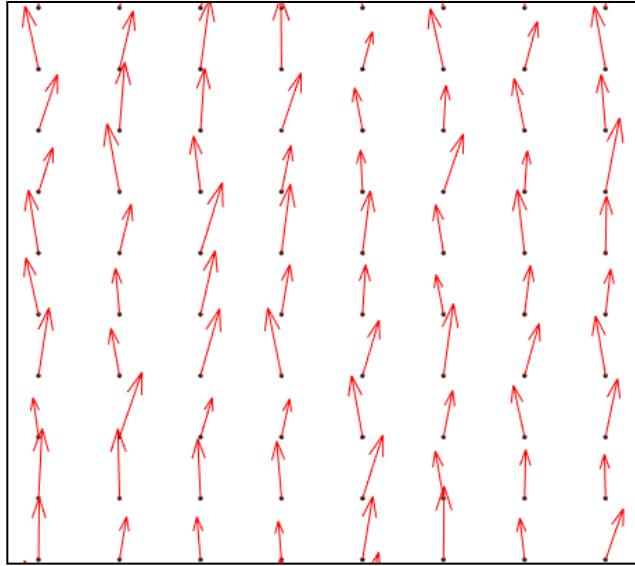


Figure 5.1: Example of shifts per grid point (black dots) with the direction of the arrow indicating migration direction and the length of the arrow the migration rate (for illustration purposes, the grid resolution and scale of migration rates is not indicated in the figure).

#### 5.1.1.3 Extrapolation of cumulative seabed levels

The second approach for extrapolation consists of extrapolating the full bandwidth of shifts in x and y directions over a specific period. This is relevant when the lower and upper envelope of predicted seabed levels is requested over a specific period. For example, the envelope for the full lifetime of the wind farm is relevant when determining initial cable burial depths.

In this approach, a polygon is drawn for each grid point which covers the area between the lower and upper bound migration direction and between the analysed grid point and maximum displacement away from the grid point (calculated by multiplying the upper bound migration rate and the number of years for which the extrapolation is performed). This polygon describes the area where the analysed grid point (thick black dot) can be during the period assessed (e.g. 2024 to 2064).

Examples for this approach are shown in Figure 5.2, with the drawn polygon shaded blue and the analysed grid point as a thick black dot. This approach is performed for all grid points resulting in the lower or upper envelope of the predicted seabed levels for the period assessed.

Note that this second approach for the extrapolation improves upon the extrapolation presented in Deltares (2020). In Deltares (2020) and Deltares (2023) the extrapolation was done using nine scenarios, corresponding to a combination between the mean, lower bound and upper bound of the migration direction and rate. In the updated methodology, all scenarios within the lower and upper bound are considered in the extrapolation.

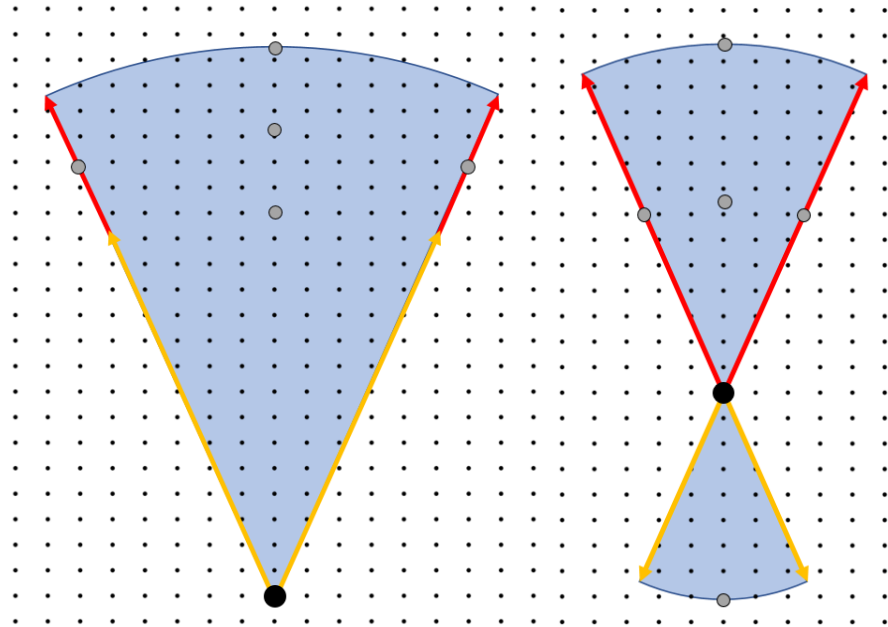


Figure 5.2: Example of the extrapolation approach indicating a case where both the lower and upper bound migration direction have the same sign (left panel) and the case where opposite migration is observed (right panel). The analysed grid point is indicated by a thick black dot whereas the other grid points are indicated by smaller black dots. The lower bound displacement for the lower and upper bound migration directions are indicated with orange arrows, the upper bound displacements by the red arrows. The area between the analysed grid point and the upper bound displacement (left panel) and between the lower and upper bound displacements (right panel) between the lower and upper bound migration directions describe where the analysed grid point can be assessed over the considered period (e.g. over the lifetime of the wind farm). All other scenarios, e.g. mean direction and mean displacement, are indicated by grey dots and always fall inside the blue polygon.

## 5.1.2 Sources of uncertainties

In assessing the future seabed levels, various sources of uncertainty have to be taken into account, both upward and downward, i.e. upward uncertainties are applied to the upper envelopes of extrapolated seabed levels, whereas the downward uncertainties are applied to the lower envelopes of extrapolated seabed levels. For the purpose of identifying depth of burial of (CE) objects only the downward uncertainty is relevant. The primary sources of uncertainty in a data-driven morphological analysis based on measured bathymetrical data are due to:

- i) Data collection and differences in the collection of data
- ii) The finite and limited grid resolution
- iii) Sand wave reshaping
- iv) Megaripple dimensions
- v) Vertical seabed level trends

These sources of uncertainty are described in more detail below.

### 5.1.2.1 i) Uncertainty due to data collection and differences in the collection of data

In the previous studies (Deltares 2020, Deltares 2023) an uncertainty of 10 centimetres was used as this was reported by Fugro (2019) for the 2018-2019 bathymetry survey. For the current analysis the 2024 survey was used as a starting point for the prediction of future seabed levels. Survey inaccuracies are not explicitly mentioned in Fugro (2025). Nevertheless, to maintain consistency with the previous analysis a value of 10 cm is again included in both the upward and downward uncertainty band for future seabed levels.

#### 5.1.2.2 ii) Uncertainty due to the finite and limited grid resolution

The bathymetrical data has a finite resolution, therefore it is unlikely that the crests and troughs of the mobile bedforms (sand waves and megaripples) are fully captured by the grid. To account for the loss in height a value of 0.10 m is included in the uncertainty band (upward), while for a loss in depth a value of 0.05 m is included (downward). Differences between values for the upward and downward uncertainty band result from the sand wave shapes observed. Sand waves have a more pronounced crest in relation to the sand wave trough. Therefore, it is expected that the loss in height for the sand wave crest is higher than the loss in depth.

#### 5.1.2.3 iii) Uncertainty due to sand wave reshaping

To account for changes in the shape of the sand waves, an uncertainty of 0.25 m at the crest and 0.15 m at the rest of the sand wave is applied. This value is based on expert judgement and results from the assessments for IJmuiden Ver and Hollandse Kust (west) (Deltares 2020, Deltares 2023).

#### 5.1.2.4 vi) Uncertainty in megaripple dimensions

For smaller-scale bedforms for which no historical trends could be determined, the inclusion in the extrapolation of seabed levels is treated in three parts. This is when bedforms are highly transient or subject to changes under extreme events.

First, the spatial distribution of megaripples is included in predicting future bathymetries as the unfiltered mobile bathymetry is used in the extrapolation. Second, the uncertainty induced from using a coarser grid in the analysis compared to the resolution of the measured data is quantified and included as described under ii).

Lastly, a significant limitation is related to the temporal variation of the megaripples. The bathymetrical surveys are a snapshot recording at a single point, while the occurrence and dimensions of the smaller-scale bedforms such as megaripples may vary in time. Therefore, a fixed value is taken up in the uncertainty band to cover this temporal variation. For the determination of the seabed levels an uncertainty of 0.10 m is used in both the upward and downward direction.

#### 5.1.2.5 Uncertainty in vertical seabed level trends

Vertical seabed level trends can be determined for the large-scale bed levels (for these layers a more sinusoidal vertical trend is expected). This results in a yearly vertical trend which can be extrapolated. However, given uncertainties or offsets in vertical seabed levels, especially for the older bathymetry surveys, these yearly trends are generally uncertain. Instead, a yearly uncertainty is added to the extrapolation of seabed levels. This uncertainty is applied uniformly over the area of interest both in downward and upward direction. Here, an additional uncertainty of 0.5 cm/year is included. This uncertainty becomes larger as the prediction time increases, resembling slow vertical trends in the large-scale bathymetry.

Note that the uncertainty in vertical seabed levels trends was not included in the methodology and results for Deltares (2020) and Deltares (2023). The range between the lowest and highest estimated seabed levels determined in this study might therefore be slightly larger compared to the previous studies as it accounts for this additional uncertainty.

### 5.1.2.6 Summary of uncertainties

A summary of the uncertainties applied to the extrapolated bed level is provided in Table 5.1.

Table 5.1: Summary of the uncertainties applied to the extrapolated future seabed levels.

| Source of uncertainty                                     | Upward [m]                           | Downward                             |
|-----------------------------------------------------------|--------------------------------------|--------------------------------------|
| Data collection and differences in the collection of data | 0.10 m                               | 0.10 m                               |
| The finite and limited grid resolution                    | 0.10 m                               | 0.05 m                               |
| Sand wave reshaping                                       | 0.15 – 0.25 m<br>(spatially variant) | 0.15 – 0.25 m<br>(spatially variant) |
| Megaripple dimensions                                     | 0.10 m                               | 0.10 m                               |
| Vertical seabed level trends                              | 0.5 cm/year<br>(temporally variant)  | 0.5 cm/year<br>(temporally variant)  |

### 5.1.3 Future seabed level assessment

The measured bathymetry, results for the extrapolation approach and uncertainties are combined to determine the future seabed levels. The method consists of the following components, in which the component for smaller-scale bedforms such as megaripples is taken up in the sand waves and uncertainties:

- Large-scale seabed;
- Sand wave (including megaripples) field; and
- Uncertainties.

An example of such an extrapolation is shown in Figure 5.3, indicating bed level predictions including bandwidths of different contributors. Note that this figure shows the seabed levels at a fixed location over time. The figure presents sand wave variability concerning the underlying large-scale seabed. The upward and downward variability due to uncertainties is included concerning the sand wave variability. The influence of future infrastructure, such as foundations and scour protections, is not considered in the presented bandwidths because of uncertainties. This influence can, however, be significant on local seabed level variations.

The evolution over time of both the lowest and highest seabed levels is not a linear process. A given location can expect both lowering and rising of the seabed over a considered period in case a sand wave passes that location. In the example shown in Figure 5.3, the highest seabed level remains relatively constant during the initial period (2020–2029), as the crest of a sand wave gradually approaches the location. Around 2030, the crest passes over the site, resulting in an increase in the highest recorded seabed level. From 2031 to 2047, this level remains unchanged, indicating that no higher sand wave crests pass during that time. A new highest seabed level is recorded between 2048 and 2051, when a larger sand wave migrates over the location.

In this analysis, future seabed levels are determined annually for 2024 to 2027, and at five-year intervals for 2029 to 2064. Each of the future seabed levels has several components, including seabed trends and uncertainties. The lowest and highest seabed levels are expressed as cumulative, being the upper and lower envelope of seabed predictions over a given period.

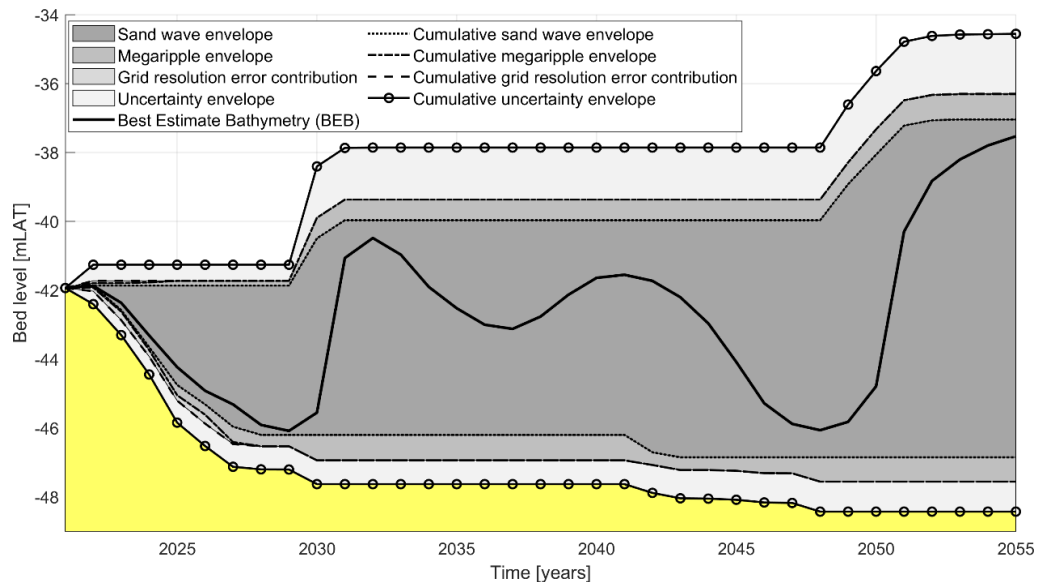


Figure 5.3: Example of future bed level predictions (i.e. bed level as function of the time) for a fixed location including bandwidths of different contributors. Note that the predictions are valid for a single exact location and do not represent an average over a defined area. The fixed location shown in this figure is not located within the area of interest analysed in this report.

The following future seabed levels for various years and periods are considered:

- Best-Estimate Bathymetry (BEB): the best-estimate of future seabed levels for a specific year determined by combining the large-scale bathymetry with the best estimate sand wave field prediction;
- Lowest SeaBed Level (LSBL): the predicted lowest minimum seabed level for a specific period (indicated by the lower envelope presented in Figure 5.3) determined by combining the large-scale bathymetry with the lowest estimate sand wave field prediction including downward uncertainty;
- Highest SeaBed Level (HSBL): the predicted highest minimum seabed level for a specific period (indicated by the upper envelope presented in Figure 5.3) determined by combining the large-scale bathymetry with the highest estimate sand wave field prediction including upward uncertainty.

The BEB provides a rough indication of the possible seabed development during the lifetime of the wind farm and should not be treated as a firm design parameter. For design, the LSBL and HSBL provide better information (maximum expected potential seabed level variations at each location during the considered period). However, the BEB does provide a valuable estimate of the seabed to compute the most probable O&M costs (e.g. related to expected cable re-burial length).

The predicted seabed level changes presented in this study follow from the applied morphological analysis techniques, describing the (uncertainty of the) physics and the natural variability of the analysed morphological system. The influence of human interventions is not considered in this analysis. Moreover, no additional safety margins for design purposes have been applied. The seabed near possible scour protections and cable crossings will most likely lower more than the LSBL levels because of local scour and edge scour.

## 5.2 Results

This report includes figures for the 2064 BEB, along with the LSB and HSBL changes between 2024 and 2064. Data and figures of seabed level projections for the other years (annually for 2024 to 2027, and at five-year intervals for 2029 to 2064) are available in the accompanying data package. An overview of the files in the data package is provided in Appendix A to C.

### 5.2.1 Best Estimate Bathymetry

The BEB is determined by combining the large-scale bathymetry with the best estimate sand wave field prediction. Figure 5.4 presents the BEB for the year 2064, and the seabed level change (lowering and rising) as the difference between the 2064 BEB and the 2024 bathymetry. Overall, the bathymetric patterns in the 2064 BEB closely resemble those of 2024. However, notable differences emerge when examining the seabed level changes. Bed level rises of up to 3 meters are observed along the steeper slopes of sand waves, while lowering between 1 and 2 meters occurs along the gentler slopes. These bed level changes are directly related to the sand wave dimensions and migration directions.

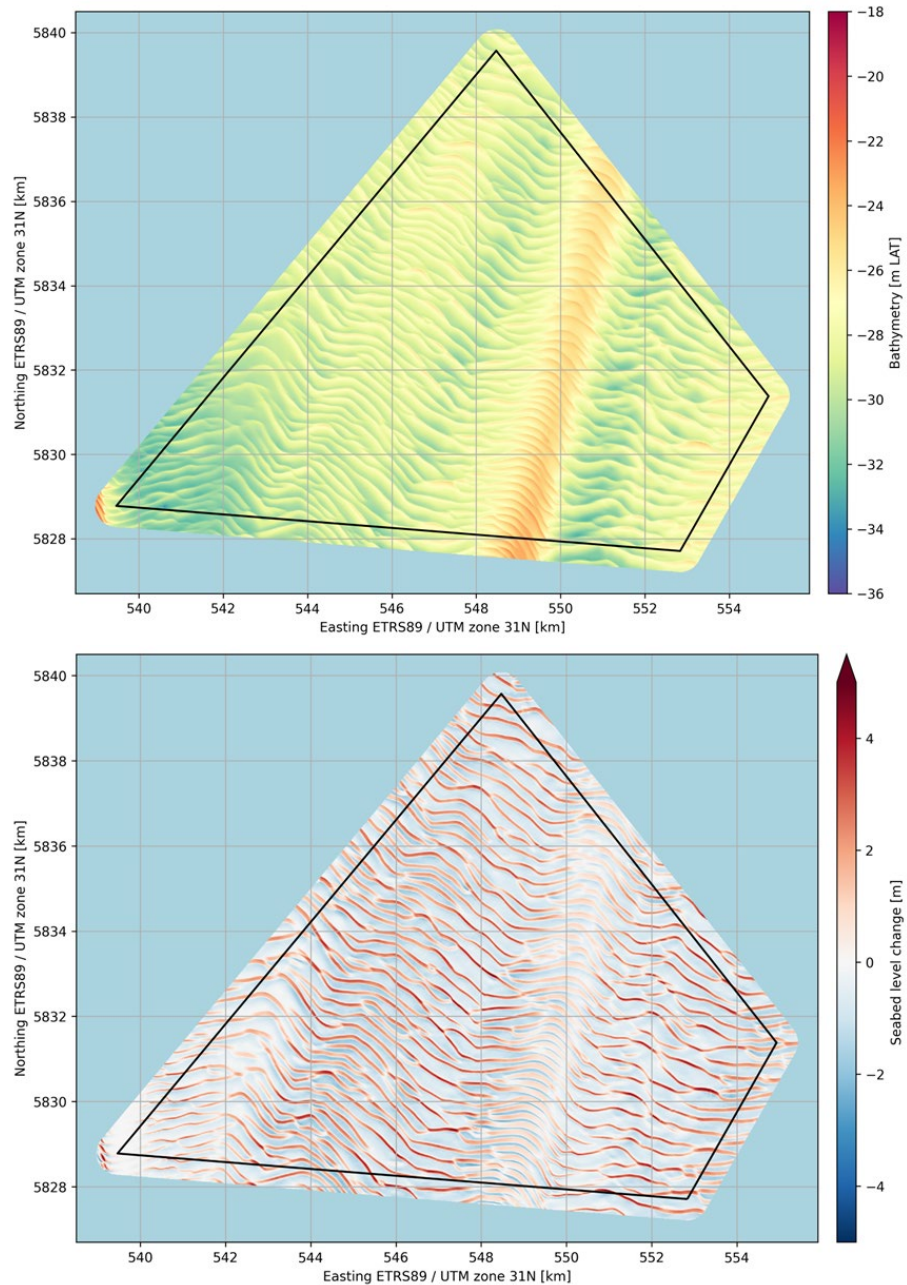


Figure 5.4: BEB for the year 2064 (top), and seabed level changes (lowering and rising) calculated as the difference between the 2064 BEB and the 2024 bathymetry (bottom).

### 5.2.2 Lowest SeaBed Level

The LSBL is estimated by combining the large-scale bathymetry, lower envelope of the sand wave field and the downward uncertainty. Figure 5.5 presents the LSBL for the period 2024 to 2064, and the seabed level change (lowering and rising) as the difference between the 2064 LSBL and the 2024 bathymetry. While the overall bathymetric patterns in the 2064 LSBL remain similar to those of 2024, bed levels are generally deeper by 1 to 2.5 metres, with local decreases reaching as much as 3.5 metres, and with less pronounced sand wave crests.

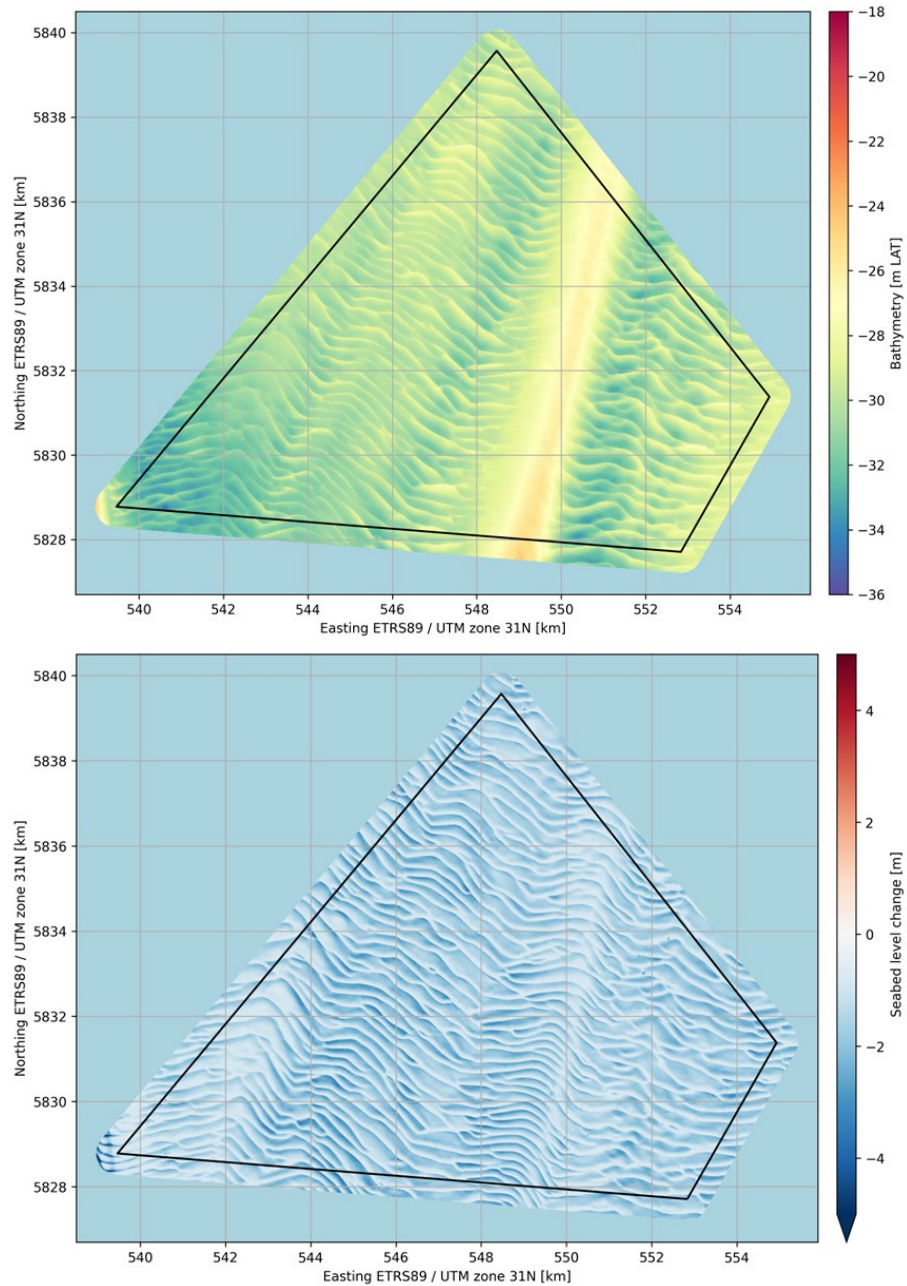


Figure 5.5: LSBL for the period 2024 to 2064 (top), and seabed level changes (lowering and rising) calculated as the difference between the 2064 LSBL and the 2024 bathymetry (bottom).

### 5.2.3 Highest SeaBed Level

The HSBL is estimated by combining the large-scale bathymetry, upper envelope of the sand wave field and the upwards uncertainty. Figure 5.6 presents the HSBL for the period 2024 to 2064, and the seabed level change (lowering and rising) as the difference between the 2064 HSBL and the 2024 bathymetry. As with the LSBL, the overall bathymetric patterns in the 2064 HSBL closely resemble those of the 2024 bathymetry. Bed levels are generally around 1.5 metres higher, with local increases reaching as much as 5 metres, and sand wave troughs appear less pronounced.

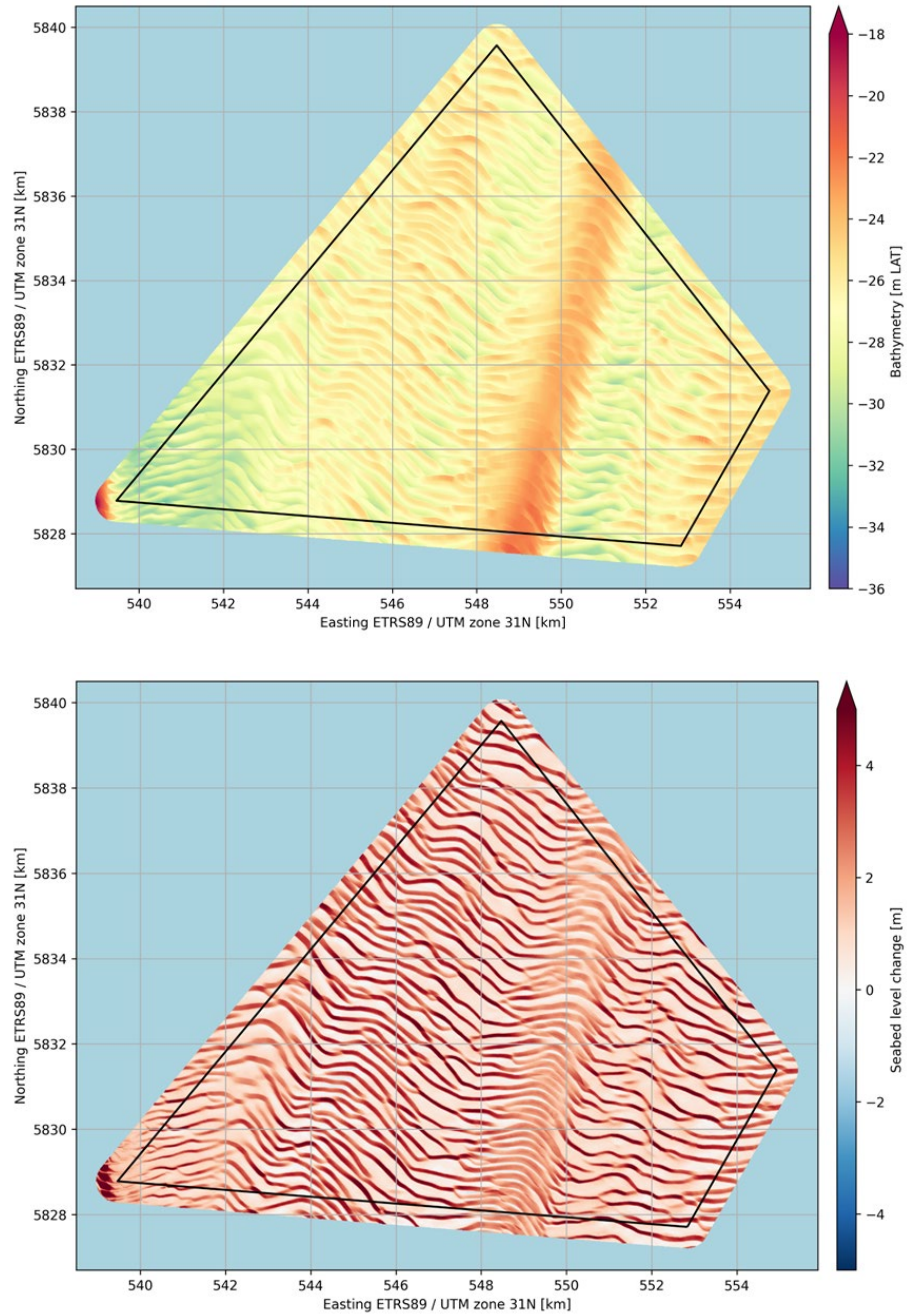


Figure 5.6: HSBL for the period 2024 to 2064 (top), and seabed level changes (lowering and rising) calculated as the difference between the 2064 HSBL and the 2024 bathymetry (bottom).

## 6 Conclusions

JBO (Jörss - Blunck - Ordemann) GmbH has requested Deltares to update the morphodynamic assessment for the HKW VII Wind Farm Site, following the acquisition of an additional 2024 bathymetric survey for the area. The objective of this update is twofold: (1) validate the results of the previous study (Deltares 2023) with the newly acquired bathymetry survey; and (2) to update the methodology, originally detailed in Deltares (2020), and results by incorporating this new survey.

### 1. Validation of the previous study

The comparison between the predicted 2024 bathymetry (Deltares 2023) and measured 2024 bathymetry shows generally good agreement, with most differences attributable to survey inconsistencies or localised human interventions. The Root Mean Square Error (RMSE) of 0.19 m falls within acceptable survey accuracy limits, and the measured bathymetry largely lies within the predicted uncertainty bounds. These results confirm that the applied methodology is suitable for predicting seabed levels in the area of interest.

### 2. Update the methodology and results

The bathymetry in HKW VI has a non-uniform morphology consisting of a prominent sand bank and a sand wave field that covers the full area. Sand waves in the area have heights between 1-5 m and wavelengths of 100-500 m. Larger sand waves are found west of the sand bank, while smaller ones appear atop it. Sand wave migration rates range from 1-4 m/year, generally moving north-northeast, influenced by the underlying large-scale bathymetry.

Future seabed levels at HKW VI were assessed by combining the large-scale bathymetry, sand wave predictions (including megaripples), and associated uncertainties. Projections from 2024 to 2064 show that seabed changes are driven by sand wave migration, with both rising and lowering trends depending on the location. The Best-Estimate Bathymetry (BEB) gives a likely scenario for seabed development, while the Lowest and Highest SeaBed Levels (LSBL and HSBL) provide conservative bounds for design purposes. The 2064 BEB shows seabed rises up to 3 m around steep sand wave slopes and lowering of 1-2 m on gentler slopes. The LSBL and HSBL for 2024 to 2064 predict decreases between 1-2.5 m (locally up to 3.5) and increases around 1.5 m (locally up to 5 m), respectively.

The predicted seabed level changes presented in this study follow from the applied morphological analysis techniques, describing the (uncertainty of the) physics and the natural variability of the analysed morphological system. The influence of human interventions is not considered in this analysis. Moreover, no additional safety margins for design purposes have been applied. The seabed near possible scour protections and cable crossings will most likely lower more than the LSBL levels because of local scour and edge scour.

Data and figures of seabed level projections for 2024 to 2064 (annually for 2024 to 2027, and at five-year intervals for 2029 to 2064) are available in the accompanying data package. An overview of the files in the data package is provided in Appendix A to C.

# References

Arcadis Germany GmbH and Geo-Engineering.org GmbH (2018). Hollandse Kust (west) Wind Farm Zone Geological Desk study. Ref: 180017; final report, dated 28 May 2018.

Damen, J. M., et al. (2018). "Spatially varying environmental properties controlling observed sand wave morphology." Journal of Geophysical Research: Earth Surface **123**(2): 262-280.

Deltares (2011). The scientific validation of the hydrographic survey policy of the Netherlands Hydrographic Office, Royal Netherlands Navy. Ref: 1201907-000-BGS-0008; final report data October 2011.

Deltares (2017). "Quickscan wind zones Hollandse Kust West 2-4. Ref: 11200745; memo, dated 24 February 2017."

Deltares (2020). Morphodynamics for Hollandse Kust (west) Wind Farm Zone. Ref: 11204811-002-HYE-0001; final report, dated 6 July 2020.

Deltares (2023). Hollandse Kust West additional morphodynamic data. Ref: 11209669-002-HYE-0001, ref 1.0. Dated 23 August 2023.

Deltares (2023). Morphodynamics for IJmuiden Ver Wind Farm Zone. WOZ2220017. Ref: 11208404-002-HYE-0001.

DHI (2019). Metocean desk study and database for Dutch Wind Farm Zones Feasibility level study for IJmuiden-Ver, Ten Noorden van de Waddeneilanden & Hollandse Kust (west). Ref: 11822658; final report, dated: 12 March 2019. Retrieved from: <https://offshorewind.rvo.nl/file/view/55039978/Metocean+Study+%28Feasibility+report%29>.

Fugro (2019). Geophysical Results Report Hollandse Kust (West) Wind Farm Zone Survey 2018. Ref: P904162; final report, dated 19 August 2019.

Fugro (2025). Geophysical Results Report  
RWE NL HKW VII UXO survey | Netherlands, North Sea  
F241863-REP-013 | 01 | 18 February 2025.

International Hydrographic Organization (2020). Standards for Hydrographic Surveys 6th edition. IHO publication No. 44.

Royal Netherlands Navy - Hydrographic Office (1980-2021). Online availability of data of the Netherlands Hydrographic Service.

Van Dijk, T. A. G. P., C. Van der Tak, W.P. De Boer, M.H.P. Kleuskens, P.J. Doornenbal, R.P. Noorlandt and V.C. Marges. (2011). "The scientific validation of the hydrographic survey policy of the Netherlands Hydrographic Office, Royal Netherlands Navy." Deltares, Report 1201907-000-BGS-0008: 165 pp.

## A Seabed levels at WTGs

This appendix consists of a table which contains the estimated seabed levels (BEB, HSBL and LSBL) for 2027 and 2064 at the WTG locations (Table A.1). This table is a subset of the excel provided in the accompanying data package which contains seabed levels annually for 2024 to 2027, and at five-year intervals for 2029 to 2064. Moreover, the data package also provides these seabed levels in a spatial data format (i.e. as a geojson file).

Table A.1: Estimated seabed levels (BEB, HSBL and LSBL) for 2027 and 2064 at the WTG locations.

| Name   | Easting  | Northing | BEB<br>2027<br>[m<br>LAT] | HSBL<br>2024<br>2027<br>[m<br>LAT] | LSBL<br>2024<br>2027<br>[m<br>LAT] | BEB<br>2064<br>[m<br>LAT] | HSBL<br>2024<br>2064<br>[m<br>LAT] | LSBL<br>2024<br>2064<br>[m<br>LAT] |
|--------|----------|----------|---------------------------|------------------------------------|------------------------------------|---------------------------|------------------------------------|------------------------------------|
| HWB 01 | 548259   | 5839131  | -28.03                    | -27.4                              | -28.46                             | -28.75                    | -27.03                             | -29.73                             |
| HWB 02 | 549064.9 | 5838631  | -29.28                    | -28.71                             | -29.92                             | -27.71                    | -26.2                              | -30.25                             |
| HWB 03 | 547139   | 5837774  | -28.19                    | -27.46                             | -28.64                             | -29.17                    | -27.06                             | -30.01                             |
| HWB 04 | 548373   | 5837611  | -27.89                    | -27.29                             | -28.27                             | -28.76                    | -26.86                             | -29.61                             |
| HWB 05 | 549822.6 | 5837670  | -29.42                    | -28.82                             | -29.81                             | -30.2                     | -28.41                             | -31.11                             |
| HWB 06 | 545958   | 5836374  | -29.19                    | -28.44                             | -29.61                             | -27.81                    | -26.68                             | -30.04                             |
| HWB 07 | 547139   | 5836553  | -28.39                    | -27.75                             | -28.85                             | -26.03                    | -24.99                             | -29.46                             |
| HWB 08 | 548970   | 5836455  | -30.16                    | -29.63                             | -30.55                             | -30.72                    | -29.21                             | -31.54                             |
| HWB 09 | 550636   | 5836638  | -26.77                    | -26.17                             | -27.25                             | -26.2                     | -24.17                             | -27.57                             |
| HWB 10 | 545087   | 5835331  | -28.14                    | -27.45                             | -28.56                             | -29.19                    | -27.06                             | -30.34                             |
| HWB 11 | 547939   | 5835331  | -27.82                    | -26.95                             | -29.22                             | -28.4                     | -26.43                             | -29.76                             |
| HWB 12 | 549510   | 5835010  | -27.15                    | -26.57                             | -27.6                              | -27.91                    | -26.18                             | -28.97                             |
| HWB 13 | 551286   | 5835813  | -25.62                    | -24.91                             | -26.05                             | -26.26                    | -24.52                             | -27.14                             |
| HWB 14 | 551945   | 5834978  | -30.09                    | -29.16                             | -31.07                             | -29.03                    | -27.49                             | -31.39                             |
| HWB 15 | 544199   | 5834267  | -28.19                    | -27.48                             | -28.64                             | -29.13                    | -27.09                             | -30.18                             |
| HWB 16 | 546107   | 5834710  | -28.15                    | -27.44                             | -28.55                             | -28.98                    | -26.81                             | -29.85                             |

|                   |          |         |        |        |        |        |        |        |
|-------------------|----------|---------|--------|--------|--------|--------|--------|--------|
| <b>HWB<br/>17</b> | 547838   | 5833978 | -26.68 | -26.01 | -27.1  | -27.56 | -25.62 | -28.67 |
| <b>HWB<br/>18</b> | 550860   | 5834214 | -25.39 | -24.77 | -25.81 | -25.5  | -23.59 | -27.06 |
| <b>HWB<br/>19</b> | 552596.5 | 5834152 | -30.54 | -29.95 | -30.95 | -27.27 | -26.05 | -31.37 |
| <b>HWB<br/>20</b> | 543317   | 5833210 | -28.94 | -28.36 | -29.35 | -29.35 | -28.03 | -30.15 |
| <b>HWB<br/>21</b> | 544866   | 5833320 | -29.07 | -28.49 | -29.48 | -29.69 | -26.66 | -30.76 |
| <b>HWB<br/>22</b> | 546475   | 5833425 | -28.98 | -28.38 | -29.36 | -29.57 | -27.89 | -30.38 |
| <b>HWB<br/>23</b> | 549090   | 5833328 | -28.77 | -28.14 | -29.22 | -26.93 | -25.34 | -29.98 |
| <b>HWB<br/>24</b> | 550742   | 5832944 | -26.98 | -26.33 | -27.41 | -24.66 | -23.46 | -27.98 |
| <b>HWB<br/>25</b> | 553256   | 5833315 | -30.02 | -29.49 | -30.41 | -27.41 | -24.94 | -30.89 |
| <b>HWB<br/>26</b> | 542448.4 | 5832170 | -30.11 | -29.55 | -30.54 | -30.27 | -29.25 | -31.08 |
| <b>HWB<br/>27</b> | 545604   | 5831956 | -29.3  | -28.61 | -29.75 | -27.63 | -26.37 | -30.44 |
| <b>HWB<br/>28</b> | 547139   | 5832070 | -30.35 | -29.78 | -30.74 | -30.42 | -26.72 | -31.4  |
| <b>HWB<br/>29</b> | 549072   | 5832115 | -27.81 | -27.11 | -28.2  | -28.68 | -26.7  | -29.48 |
| <b>HWB<br/>30</b> | 551904   | 5832625 | -28.38 | -27.71 | -28.85 | -29.32 | -27.32 | -30.68 |
| <b>HWB<br/>31</b> | 553981.8 | 5832395 | -27.54 | -26.93 | -27.95 | -28.68 | -26.66 | -29.87 |
| <b>HWB<br/>32</b> | 541649   | 5831085 | -30.1  | -29.51 | -30.62 | -30.67 | -29.05 | -31.64 |
| <b>HWB<br/>33</b> | 543542   | 5831504 | -29.96 | -29.39 | -30.36 | -30.08 | -28.52 | -30.87 |
| <b>HWB<br/>34</b> | 544771   | 5830897 | -29.78 | -29.09 | -30.19 | -27.52 | -26.39 | -31.02 |
| <b>HWB<br/>35</b> | 547914   | 5830920 | -29.06 | -28.2  | -29.97 | -27.5  | -25.73 | -30.29 |
| <b>HWB<br/>36</b> | 550195.6 | 5831495 | -25.72 | -25.14 | -26.15 | -25.58 | -24.39 | -26.6  |
| <b>HWB<br/>37</b> | 552460   | 5831467 | -27.5  | -26.84 | -27.99 | -28.51 | -26.45 | -29.71 |
| <b>HWB<br/>38</b> | 554736.4 | 5831438 | -28.49 | -27.82 | -28.91 | -26.68 | -24.98 | -29.25 |
| <b>HWB<br/>39</b> | 540678   | 5830049 | -32.32 | -31.8  | -32.75 | -31.13 | -28.97 | -33.1  |

|                   |        |         |        |        |        |        |        |        |
|-------------------|--------|---------|--------|--------|--------|--------|--------|--------|
| <b>HWB<br/>40</b> | 542050 | 5830077 | -30.78 | -30.22 | -31.29 | -29.97 | -28.69 | -31.61 |
| <b>HWB<br/>41</b> | 543642 | 5829870 | -27.79 | -27.09 | -28.23 | -28.74 | -26.7  | -29.72 |
| <b>HWB<br/>42</b> | 546178 | 5830370 | -27.44 | -26.58 | -27.9  | -29.18 | -26.19 | -30.37 |
| <b>HWB<br/>43</b> | 549431 | 5830008 | -24    | -23.41 | -25.21 | -25.69 | -22.74 | -26.62 |
| <b>HWB<br/>44</b> | 551070 | 5830387 | -30.44 | -29.79 | -30.92 | -30.18 | -29.21 | -31.23 |
| <b>HWB<br/>45</b> | 552715 | 5830476 | -31.04 | -30.45 | -31.44 | -31.2  | -29.08 | -32.14 |
| <b>HWB<br/>46</b> | 553924 | 5830404 | -27    | -26.32 | -27.42 | -27.65 | -25.94 | -28.67 |
| <b>HWB<br/>47</b> | 546337 | 5829060 | -29.69 | -28.81 | -30.85 | -29.44 | -27.66 | -31.17 |
| <b>HWB<br/>48</b> | 547976 | 5829466 | -28.22 | -27.75 | -29.09 | -27.18 | -24.65 | -29.42 |
| <b>HWB<br/>49</b> | 551193 | 5828921 | -29.94 | -29.37 | -30.45 | -27.12 | -25.56 | -30.87 |
| <b>HWB<br/>50</b> | 552850 | 5829043 | -28.53 | -27.84 | -28.95 | -29.4  | -27.43 | -30.18 |
| <b>HWB<br/>51</b> | 539786 | 5828874 | -31.31 | -30.71 | -31.72 | -31.3  | -30.06 | -32.14 |
| <b>HWB<br/>52</b> | 540859 | 5828790 | -32.28 | -31.63 | -32.71 | -32.42 | -31.17 | -33.22 |
| <b>HWB<br/>53</b> | 542191 | 5828681 | -30.82 | -30.15 | -31.26 | -31.37 | -29.76 | -32.31 |
| <b>HWB<br/>54</b> | 543548 | 5828687 | -30.71 | -30.19 | -31.07 | -31.04 | -29.79 | -31.83 |
| <b>HWB<br/>55</b> | 544999 | 5828457 | -29.35 | -28.71 | -29.77 | -30.08 | -28.32 | -31.16 |
| <b>HWB<br/>56</b> | 547005 | 5828296 | -27.1  | -26.24 | -28.42 | -26.64 | -24.47 | -28.75 |
| <b>HWB<br/>57</b> | 548626 | 5828279 | -24.93 | -24.31 | -26.62 | -26.36 | -23.37 | -27.36 |
| <b>HWB<br/>58</b> | 549812 | 5828071 | -28.91 | -28.34 | -29.32 | -28.27 | -24.95 | -29.7  |
| <b>HWB<br/>59</b> | 551416 | 5827953 | -30.75 | -30.12 | -31.21 | -28.82 | -27.62 | -31.67 |
| <b>HWB<br/>60</b> | 552622 | 5827846 | -29.18 | -28.64 | -29.63 | -26.37 | -24.61 | -30.05 |

# B Point predictions at WTGs

This appendix consists of an example figure which contains the point prediction of the estimated seabed levels at HWB 01 between 2024 and 2064 (Figure B.1). The accompanying data package contains the point prediction figures for all WTG locations (i.e. 61 png files).

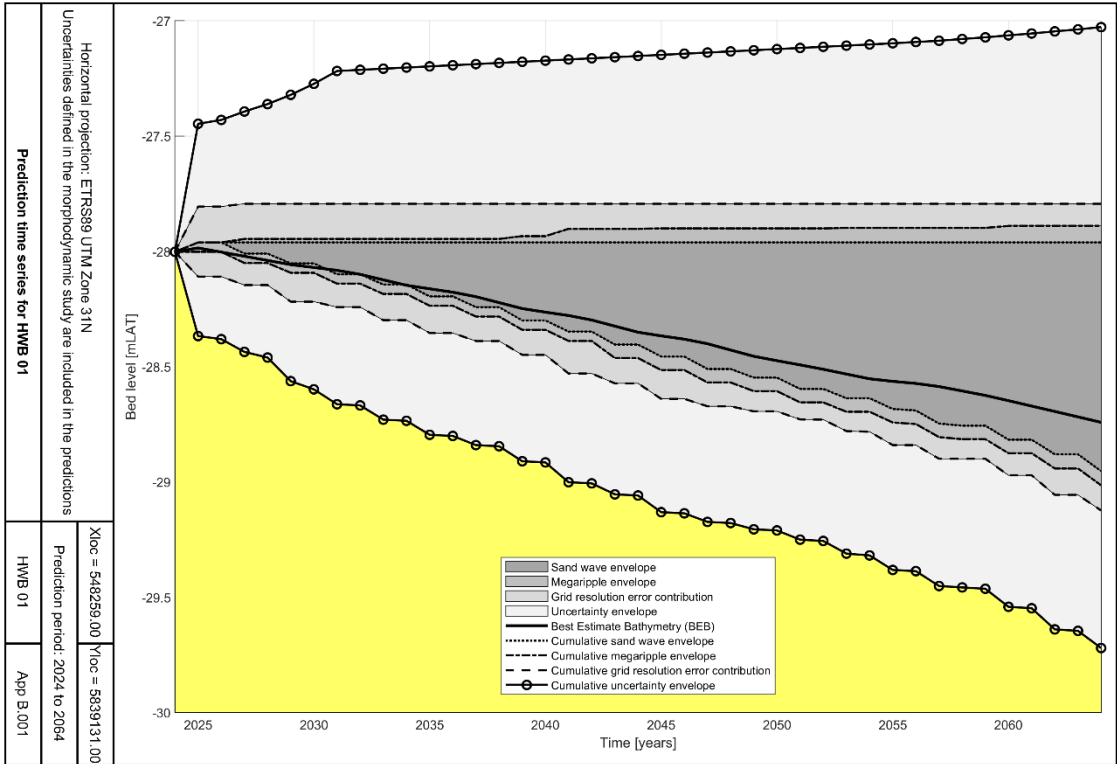


Figure B.1: Points prediction of seabed levels at HWB 01.

## C Site covering seabed level rasters

This appendix contains a figure showing the Best-Estimate Bathymetry for the year 2064. The accompanying data package contains figures and rasters (i.e. tif files) for the estimated seabeds (BEB, HSBL and LSBL) annually for 2024 to 2027, and at five-year intervals for 2029 to 2064 (i.e. 33 png files and 33 tif files).

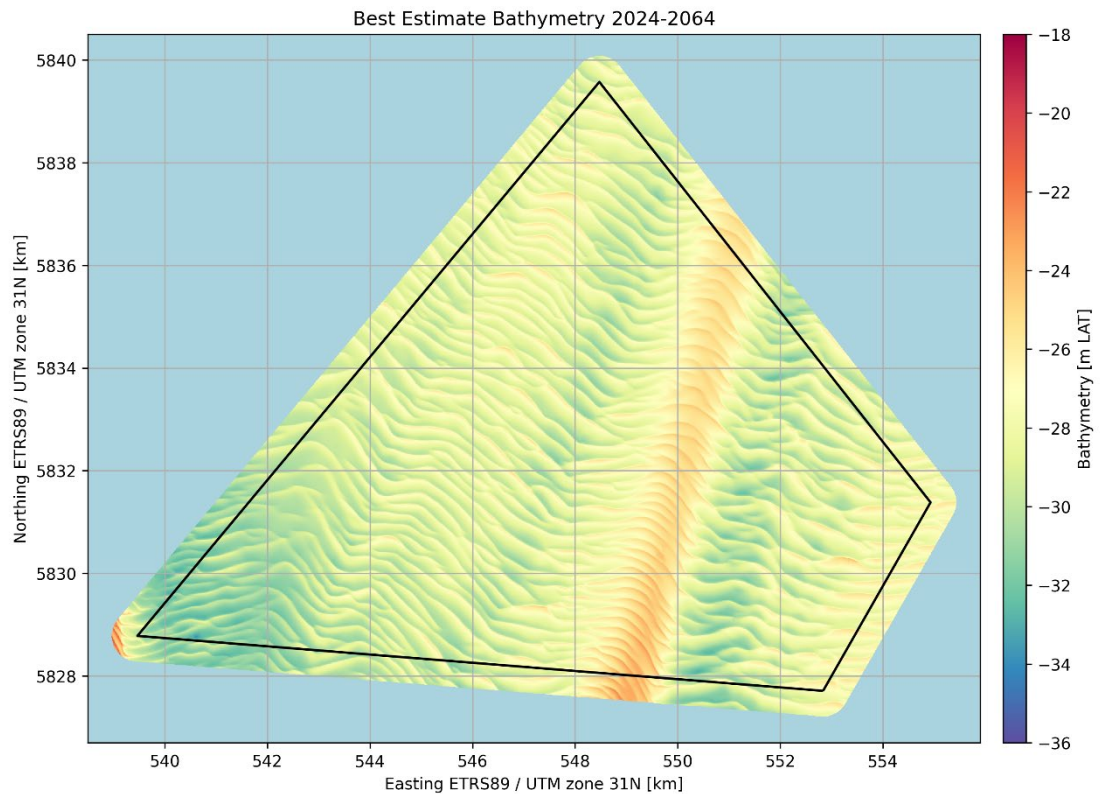


Figure C.1: Best-Estimate Bathymetry for the year 2064.

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